Juan Toro Fabra

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Education

École Polytechnique Fédérale de Lausanne (EPFL) — Bachelor in Physics 2024–2027 (expected)
- Relevant coursework: Classical Electrodynamics, Electromagnetism, Wave Physics, Physics Lab, Numerical Physics, Data Analysis, Probability & Statistics.

Universidad Autónoma de Madrid (UAM) — Grado de Física 2023–2024

ESTALMAT Madrid — Royal Academy of Exact, Physical and Natural Sciences 2017–2021
- Programme for the early detection and development of mathematical talent, with specialised courses at the Complutense University of Madrid.

Experience

Research Intern — Coastal Morphodynamics, UPC BarcelonaTech Summer 2026
- Investigated the length-scale selection governing the alongshore spacing of washover deposits during extreme storms (Prof. D. Calvete), extending the study of Carrión-Bertrán et al. (2026).
- Designed a 150+ run simulation campaign (XBeach 2DH) to test three candidate mechanisms, with paired random seeds to control for noise and control runs to rule out numerical artefacts.
- Built spectral estimators of the emergent length scale (band-integrated FFT, wavelets), quantified their uncertainty, and showed that several apparent trends were indistinguishable from noise.
- Identified discrete regime transitions that invalidate continuous-relaxation fits.
- Ran the campaign on an SGE/OpenMPI cluster and built the NetCDF post-processing pipeline in MATLAB and Python.

Teaching Assistant — University of Lausanne Spring 2026
- General Physics I and Experimental Physics II (mechanics, optics, thermodynamics): weekly exercise sessions, course forums, exam proctoring.

QMSat / SCRAP CubeSat Project Fall 2025 – Spring 2026
- Student team (EPFL, Queen Mary, Imperial College) building a CubeSat for space-debris detection in LEO, at the prototype stage.
- Lead of the Localization & Geometry software module: plate solving, orientation estimation, coordinate transformations, uncertainty modelling.

Camp Group Leader — León, Spain Summer 2025
- Sole employed leader of a group of 20 children (with one coordinator). Led the full evacuation of the group during a nearby wildfire, coordinating with local authorities.

Projects

Personal blog — juantorofabra.com 2026–Present
- Personal blog on physics, applied maths and data science (Quarto, GitHub Pages).
- Example topics: Statistical physics applied to Go, word embeddings and data science, non-mainstream physics and philosophy of physics, high-dimensional geometry.

PV & Battery Detection from Smart-Meter Data — EnergyData Hackdays 2026
- Detected rooftop PV, home batteries and EV chargers from household smart-meter time series (AEW), combining engineered features, random forests and positive–unlabeled learning.

Option Pricing Engine — Heston Model 2026
- Semi-analytical pricing via the characteristic function and calibration to market prices (Python); Monte Carlo engine in C++ with Andersen's QE scheme.

Computational Physics 2023–Present
- N-body simulation with the Barnes–Hut algorithm ($\mathcal{O}(n \log n)$, ∼50,000 bodies); Euler-equation fluid solver (stability, boundary conditions, SOR).
- Simulations and animations for social media (@JToroide): heat diffusion, Fourier series, Ising model, graph algorithms.

Skills & Languages

Programming: Python (NumPy, SciPy, Matplotlib), MATLAB, C++, LaTeX, Git, HPC (MPI, SGE)
Methods: Spectral analysis, Monte Carlo, numerical PDEs, data science.
Languages: Spanish (native), English (C2), French (C1)